Neural Activity Based on Reaction Time During Impulsive Decision Making

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Introduction

- Delay discounting (DD) tasks measure the capacity to postpone gratification. They define reward-impulsivity as a preference for sooner smaller (SS) over larger later (LL) rewards [1].
- Choices that are made more quickly also involve reward-impulsivity, i.e., fast responses executed by rapid neural processes, often focused on getting a reward, not maximizing it [2].
- Hypotheses: Fast and slow choices indicated by reaction times (RT) reflect more than valuation comparisons, they involve reward-impulsivity or evidence accumulation, and each of these is reflected in unique neural activities.

Methods

- Reward-impulsive (N=10; 4F; mean age 22.3) and non-reward-impulsive (N=10; 3F; mean age 22.5) young adults performed a DD task (average 124 trials) in an MRI scanner. **Reward-impulsivity Groups** were defined based on previous preference for SS on the same task.

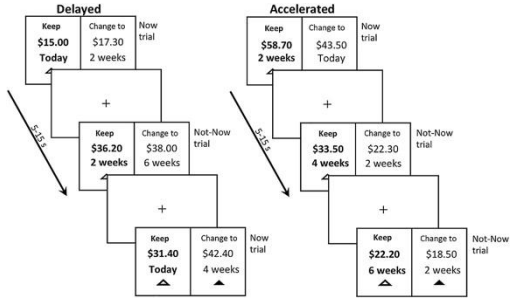


Fig. 1 DD task design with different conditions of Framing (delayed vs. accelerated), Trial Type (Now vs. Not-Now) and Reward Time Difference (2 or 4 weeks)

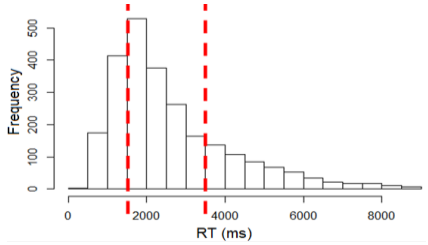


Fig. 2 RT distribution of all responses for all subject. **RTs types** are classified as: short RT (400-1500ms), medium RT (1500-3500ms) and long RT (> 3500ms).

- Behavioral Analysis: A Bayesian generalized linear mixed-effects model was performed on SS vs LL Choices as a function of SS Reward, Trial Type, Frame, Reward Time Difference, Relative Reward Difference (RRD) between SS and LL, Reward-impulsivity Group and RT (continuous).

$$Choice \approx SS\ reward + (Trial\ Type + Frame + Time\ Difference + RRD + Group) * RT + (1|sub)$$

- Imaging Analysis: Standard preprocessing (slice timing correction, motion correction, normalization and smoothing) and individual and group analysis were performed using SPM12. Individual-level analysis investigated the effect of RT Type using regressors for short, medium and long RTs with RRD as a parametric modulator. Results from individual subjects were submitted to random-effects whole brain and ROI group analyses. One-sample t-tests were used to identify differences between Choices, RT types and between Groups using whole brain and ROI analyses.

Behavioral Results

1. As expected, the likelihood of choosing SS was significantly affected by SS amount, RRD and Reward Time Difference [3].
2. The Reward-Impulsive group chose SS options more often than the Reward Non-Impulsive group (Fig. 3A).
3. The interaction of RTGroup (Fig. 3B) demonstrates that the Reward-Impulsive group made more faster SS choices than the Reward Non-Impulsive group.
4. For both groups, differences in RTs aller L

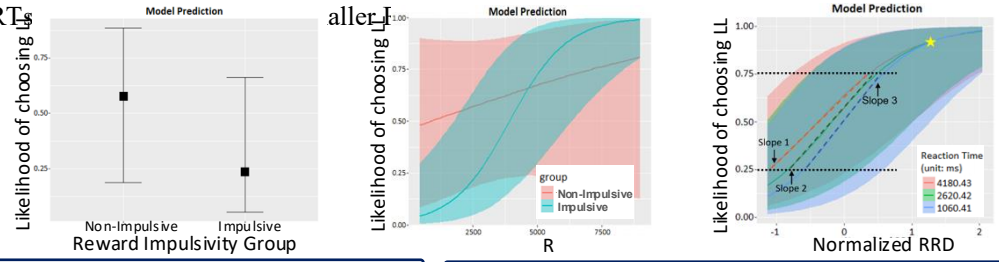


Fig 3. Model predictions of participant's choice preferences. A. Overall choice distribution for each group; B. Each group choices as a function of RT; C. choices as a function of RRD with varied RT responses.

Brain Imaging Results

1. There were no significant differences between conditions tested at whole-brain level.
2. For ROI analysis, there was significant different in BOLD activation for RT Types in the Insula, Inferior Parietal and Parahippocampus (Table 2 & Fig 4).

Region	T-test	p-value
Inferior parietal	medium RT > short RT	**0.0305**
Insula cortex	short RT> medium RT	0.0553
	short RT > long RT	**0.0020**
Parahippocampus	medium RT > short RT	**0.0151**

Table 2. Summary of ROIs analysis results.

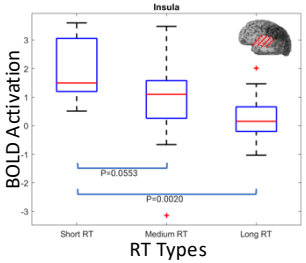


Fig 4. Insula BOLD activation for the 3 RTs.

Conclusion

- Reward-impulsivity is reflected by both the choice and the time it takes to make the choice. Reward-Impulsive subjects took longer to make LL than SS choices whereas Non-Impulsive subjects' choices were not associated with RT.
- Decision-making involved different brain regions based on participant's RT during DD task. Decisions associated with short RT shows increased activity in the network of the reward circuitry, whereas decision associated with longer RT shows increase activation in areas related to evidence accumulation and information processing.

Acknowledgement

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References

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